

CS592: Spatial Data Science

Undergrad/Graduate – Elective (CS/DS/AI)



Remote Sensing
&
Spatial Analytics

Dr. sc. Muhammad Adnan Siddique
Assistant Professor
Information Technology University - ITU



Remote Sensing & Spatial Analytics (RSA) Lab

RSA Lab aims to become a premier institution of learning and research in the domain of remote sensing and spatial analytics in Pakistan.

RSA Lab

“We tackle pressing scientific problems related to the Earth and its environment!”

Our mission is to equip members with the skills, resources, and mentorship needed to excel and lead in their fields.

Team Members



Dr. M. Adnan Siddique
Director, RSA Lab



Ehtasham Naseer
PhD Fellow, SNSF



Abdul Basit
PhD Fellow, Genoa, Italy



Murtaza Nadeem
MS thesis student



Nauman Arshad
MS thesis students



Saad Bin Hammad
MS thesis students



Izba Atif
MS thesis students

Graduated Members



Geo spatial analysis for suggestive location intelligence

Salman Bashir
ML, Engineer PITB



Ship Detection in Sentinel-1 SAR Imagery Using YOLOv8

Malik Ahmed
ML, Engineer Xeven Solutions



Land Use Land Cover Classification Using High-Resolution Aerial Imagery

Hina Amin
Looking for PhD



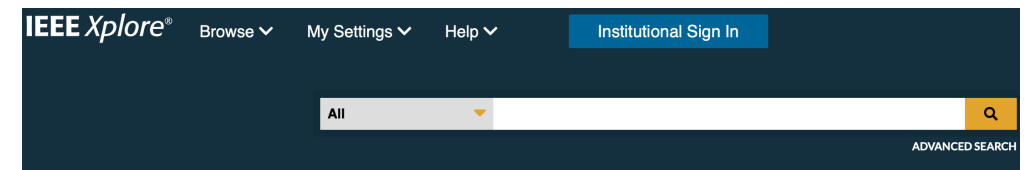
Previously in Spatial Data Science

Spring-2023



- Student projects from Spring 2023 offering were presented at an exhibition. Some of the projects won awards.
- A student from Spring 2024 published his project work in an IEEE conference.

Spring-2024



Spatio-Temporal Analysis of Annual NO₂ Concentration and Population Exposure in Palestine and Israel

Publisher: IEEE [Cite This](#) [PDF](#)

[Muhammad Nauman Arshad](#); [Malik Ahmad Javed](#); [Ehtasham Naseer](#); [Muhammad Adnan Siddique](#) [All Authors](#)





Remote Sensing & Spatial Analytics

collaborations with researchers from across the globe

ETH zürich

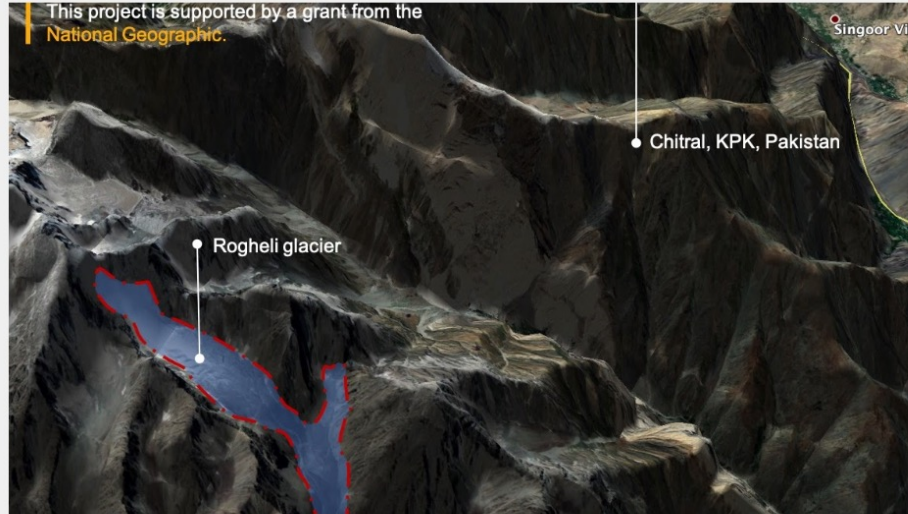


Massachusetts
Institute of
Technology

UNIVERSITY of
STIRLING



Projects



AI4GLOF

Artificial Intelligence for glacial lake outburst floods hazard potential assessment in Chitral, Pakistan

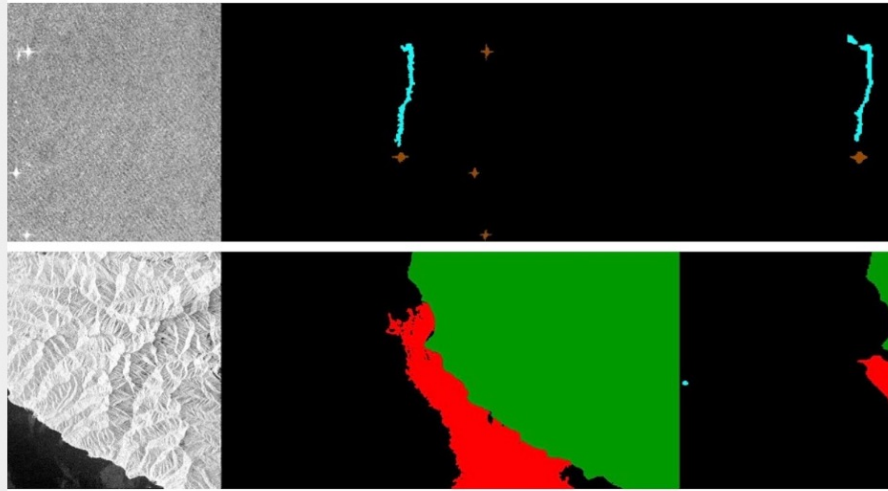
The project AI4GLOF is one of the 12 awards this year in National Geographic's "AI for Earth Innovation" program, and first-ever grant awarded to any Pakistani research team since the inception of this program.

AI4GLOF focused on the use of AI for Glacial Lake Outburst Floods hazard potential assessment in Chitral valley, Pakistan.

The Principle Investigator (PI) of the project was Dr. Adnan Siddique from the RSA Lab.



Projects



Remote-SOS4EEZ

Remote Sensing for Oil Spill Detection in Pakistan's Exclusive Economic Zone

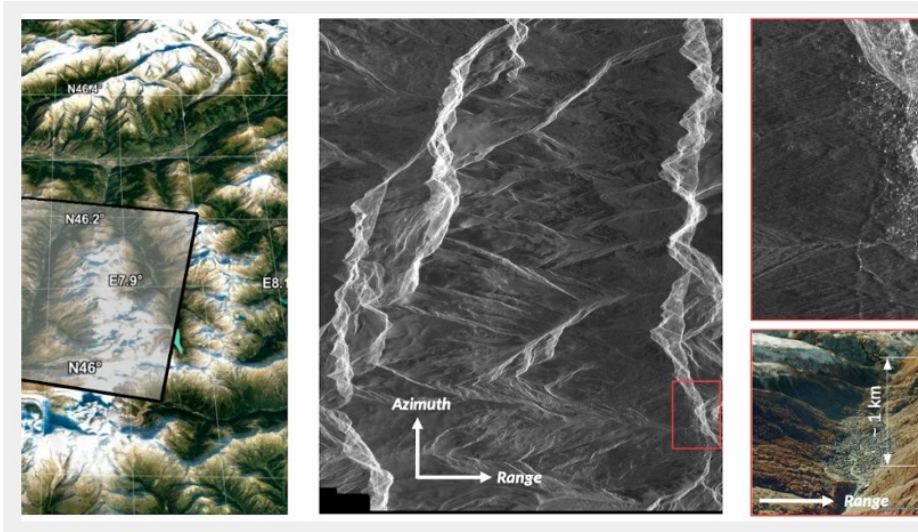
This project is supported by funding from the National Center of GIS & Space Applications (NCGSA), Government of Pakistan.

Accidental oil spills or dumping by oil tankers in the open waters is a threat to marine and coastal ecosystems. In this project, we develop a methodology based on the use of an encoder–decoder-based convolutional neural network trained on an annotated dataset comprising selected oil spill events verified by the European Maritime Safety Agency (EMSA).

The project was led by Dr. Adnan Siddique from the RSA Lab.



Projects



SPIRIT

Towards a large-scale landslide hazard assessment scheme based on time series analysis of satellite radar interferometry

This project is funded by the Swiss National Science Foundation (SNSF) SPIRIT grant IZSTZ0_216526.

This project aims at advancing SAR interferometric methods for improved identification of critical landslide and rockfall, and other mass movements in mountainous areas in general, and assessment of triggers leading to slope failures.

This project has built a transboundary research partnership between relevant experts from Switzerland and Pakistan. The project partners are: Dr. Othmar Frey (ETH Zurich), Dr. Adnan Siddique (ITU, Lahore) & Dr. Farooq Ahmed (UET, Lahore).

Mr. Ehtasham Naseer is pursuing his doctoral research within the scope of this project, under Dr. Adnan Siddique's supervision at the RSA Lab.



Course Overview

Aim

- This course aims to provides the tools, the methods, and the theory to meet the challenges of contemporary data science applied to geographic problems and data.
- But there will be a lot of programming!
- In the new world of
 - pervasive
 - large
 - frequent

data, we have new opportunities to understand and analyze the role of spatial features of data in everyday life.



Motivation

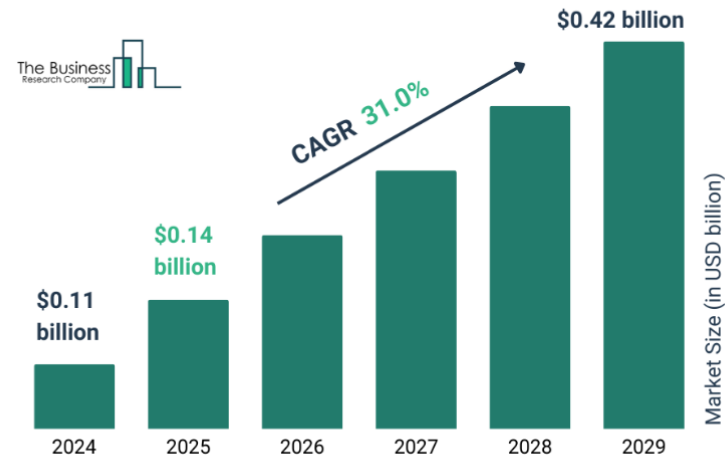
The Global Geospatial Market Size is estimated to be USD 452 billion in 2022 and is forecasted to grow at 14.61 percent CAGR and estimated to be USD 681 billion in year 2025.

However, it is expected to grow at much faster rate of 16.1 percent CAGR post 2025, reaching USD 1.44 trillion by 2030.

Source: <https://geospatialworld.net>

Geospatial Analytics AI Market Size 2025

Geospatial Analytics AI Global Market Report
2025



Source: <https://www.thebusinessresearchcompany.com>



Organization

- Credits:
 - 3 (2 lectures per week)
- Lecture Schedule:
 - Monday (04:00 – 5:30 pm) & Friday (02:30 – 04:00 pm)
- Evaluation:
 - Assignments (0%) / Quizzes (20%) / Project (30%) / Mid-Term Exam (25%) & Terminal Exam (25%)
- Instructor Email:
 - adnan.siddique@itu.edu.pk
- Visiting Hours:
 - please send an email first!
- **You are REQUIRED to carry your laptops with you in this class, with requisite libraries installed.**



Resource materials

- Lecture slides
- Readings:
 1. [RBW] Sergio J. Rey, Dani Arribas-Bel, Levi J. Wolf, Geographic Data Science with Python. Access [here](#).
 2. [CB] Lex Comber, Chris Brunsdon, Geographical Data Science and Spatial Data Analysis: An Introduction in R.
- This first part of the course will mainly follow the RBW book! The slideset is prepared accordingly.
- Copyright notice:
 - The RBW book is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License. It is freely and openly available online.
 - Any excerpts from the book remain copyrighted to the authors.



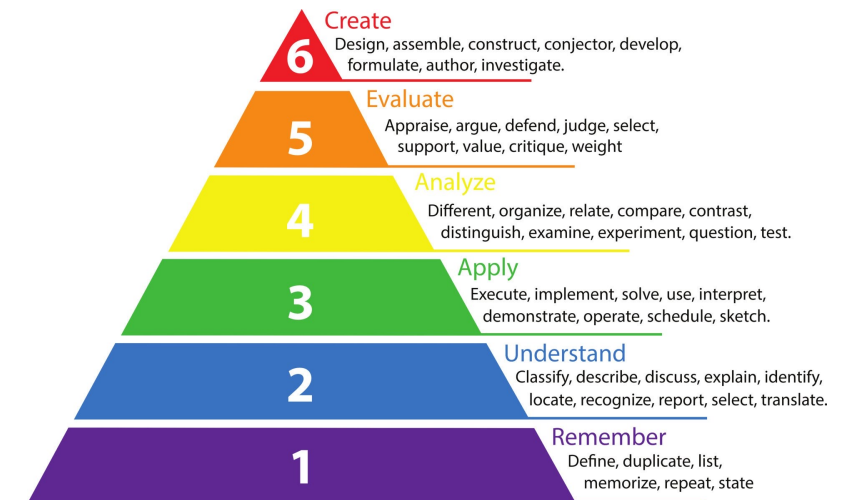
Overview

- This course is primarily about spatial » data science «
 - but it will involve some of
 - Geostatistics
 - Geocomputation, &
 - Geographic Information Systems (GIS)
 - Remote Sensing
- We'll discuss:
 - fundamentals of geographic theory for data scientists
 - distinctions between geographic models and the data structures
 - how to read/write/represent spatial data formats
 - data visualization
 - clustering, feature engineering, spatial regressions and interpolations



CLOs

CLO-GA-Bloom Mapping Matrix			
CLO#	Course Learning Outcome	Graduate Attributes (GAs)	Bloom's Taxonomy Level
CLO-1	Analyze the three fundamental spatial data structures (geographic tables, surfaces, and spatial graphs), explaining the conceptual data model each structure represents and its appropriate use case.	GA-3 (Problem Analysis)	C4 – Analyze
CLO-2	Apply the principles of Exploratory Spatial Data Analysis (ESDA) to detect, measure, and visualize geographic patterns in spatial datasets, using appropriate statistical and cartographic techniques	GA-3 (Problem Analysis)	C3 – Apply
CLO-3	Design and implement a reproducible spatial data analysis workflow to diagnose a geographic problem (e.g., inequality, clustering, spatial regression) and develop evidence-based insights.	GA-4 (Design/Development of Solutions)	C6 – Create



! Important

- The semester project has to be submitted with complete details (code, dataset), alongside a report/poster before the last week of the term. Partial evaluation of the project (10-25%) will be at the time of the midterm exams; complete evaluation will be in the last week of the term. Presentations are mandatory.
- No makeup quizzes will be offered, and all quizzes will be included for GPA computation.
- Overall, relative marking will be applied. However, projects will be marked on absolute scale.



Course materials

- Course materials and useful resources will be shared via Google Classroom.

<https://classroom.google.com/c/ODQxMTM1MTIxNjc1?cjc=h3xcc3q3>

Google Classroom

j4ht53dq

